Unit 6, Lesson 11

Partial Products and the Standard Algorithm

Standards

Addressing **4.NBT.B.5**

Instructional Routines

- Number Talk

Goals

- Calculate the product of three- or four-digit numbers and one-digit numbers.
- Compare and contrast (orally and in writing) algorithms that use partial-products, including the standard algorithm for multiplication

Student Facing Learning Goals

Let's compare multiplication algorithms.

Lesson Purpose

The purpose of this lesson is for students to analyze the standard algorithm for multiplication and compare it to an algorithm that uses partial products they saw in earlier lessons.

Narrative

In previous lessons, students analyzed and used an algorithm that uses partial products to multiply multi-digit whole numbers. They learned that an algorithm can represent the base-ten diagrams and rectangular diagrams, but it is more efficient for keeping track of and recording partial products.

This lesson extends students' analysis to include the standard algorithm for multiplication of multi-digit numbers. In IM Grade 4, the standards focus on understanding place value and how it is represented in different methods for finding products. The work here serves to build the groundwork for making sense of the standard algorithm in IM Grade 5, so students are not expected to use the standard algorithm at this time.

Access for Students with Disabilities

- Engagement

Access for English Learners

- MLR8

Lesson Timeline

Warm-up	10 min
Activity 1	20 min
Activity 2	15 min
Synthesis Estimate	10 min
Cool-down	5 min

Teacher Reflection Questions

In IM Grade 5, students will use the traditional algorithm. How does the way they analyzed two different algorithms in Activity 2 build toward this work?

Warm-up

10 min

Number Talk: The Value of the Digits

Standards

Addressing **4.NBT.B.5**

Instructional Routines

- Number Talk

This *Number Talk* routine encourages students to think about decomposing factors by place value to multiply multi-digit numbers by one-digit numbers. As students look for and make use of structure, they may notice that multiplying the ones place has a result in the pattern of 5, 10, and 15. Students may use the partial products equations to multiply.

This routine helps students pay attention to the value of the digits when multiplying. This is important for setting up the conversation about the standard algorithm, in which students will find partial products mentally and use what they know about the value of the digits to condense the number of steps to multiply by multi-digit numbers.

Student Task Statement

Find the value of each expression mentally.

- 5×101
- 5×102
- 5×203
- 5×404

Student Response

- 505: $5 \times 100 = 500$ and $5 \times 1 = 5$, and $500 + 5 = 505$.
- 510: 5×102 is 5 more than 5×101, so it is 5 more than 505, which is 510.
- 1,015: 5×200 is 1,000 and 5×3 is 15, so 5×203 is 1,015.
- 2,020: 404 is 4 times 101, 5×404 is 4 times 5×101 or 4×505, which is 2,020.

Launch

- Display one expression.
- "Give me a signal when you have an answer and can explain how you got it."
- 1 minute: quiet think time

Activity

- Record answers and strategy.
- Keep expression and work displayed.
- Repeat with each expression.

Activity Synthesis

- "What connections or relationships do you see between each expression?" (Each problem involves one more group of 5 or sets or 100 more.)

Activity 1

20 min

Two Algorithms to Multiply

Standards

Addressing **4.NBT.B.5**

This activity introduces students to the standard algorithm for multiplication. Students make sense of it by comparing and contrasting it to an algorithm that uses partial products for multiplying three- and four-digit numbers by one-digit numbers where no regrouping is necessary. When they interpret the given student work showing the

standard algorithm students construct a viable argument for what Kiran did in his calculation (MP3). They also have an opportunity to make use of the structure they notice to compute the value of other products.

Access for English Language Learners

MLR8 Discussion Supports. Synthesis: Create a visual display of Diego's algorithm. As students share their strategies, annotate the display to illustrate connections. For example, next to each step, write the multiplication expression used to find each partial product.
Advances: Speaking, Representing

Student Task Statement

1. Here are two algorithms for finding the value of 3×713.

Kiran

$$\begin{array}{r} 7\ 1\ 3 \\ \times\quad\ \ 3 \\ \hline 2,1\ 3\ 9 \end{array}$$

Diego

$$\begin{array}{r} 7\ 1\ 3 \\ \times\quad\ \ 3 \\ \hline 9 \\ 3\ 0 \\ +\ 2,1\ 0\ 0 \\ \hline 2,1\ 3\ 9 \end{array}$$

Discuss with your partner:

   a. How are these algorithms alike? How are they different?

   b. How do you think Kiran found the product 2,139?

2. Find the value of each product.

   - 212×4
   - $3 \times 4{,}132$

Student Response

1. Sample response:
   a. They both have the same result. Kiran's way doesn't show his steps, but you can see the partial products in Diego's way.
   b. Kiran was doing the same process as Diego in terms of finding partial products, but he added the products in his head after finding each partial product.
2.
   - 848
   - 12,396

Launch

- Groups of 2

Activity

- "Make sense of Kiran and Diego's algorithms. Talk to your partner about how they are alike and different, and how Kiran might have found his answer."
- 3 minutes: partner discussion
- Pause for a discussion.
- Invite students to share their conjectures on how Kiran might have reasoned about the product.
- Alternatively, consider displaying a few scenarios and polling students on which one might be closest to their conjecture, if any. For example:
  - A: Kiran found $(3 \times 700) + (3 \times 13)$ mentally and wrote down the result of $2{,}100 + 39$.
  - B: Kiran used the same method as Diego but added the 9, 30, and 2,100 mentally, without writing them down.
  - C: Kiran drew a diagram and did the computation on another sheet of paper and wrote the result here.
  - D: Kiran multiplied the single-digit 3 with each digit in 713 and wrote each partial product in a single line.
- Explain that Kiran had multiplied 3 by each digit in 713, but instead of reasoning about 3×3, 3×10, and 3×700, Kiran reasoned about 3×3, 3×1, and 3×7, while paying attention to the place value of each digit.
- Demonstrate Kiran's process:
  - Because the 3 in 713 means 3 ones, he wrote the result of 3×3 or 9 in the ones place.
  - Because the 1 means 1 ten, he wrote the result

of 3×1 in the tens place.
    - Because the 7 means 7 hundreds, he wrote the result of 3×7 in the hundreds place.
- "Try using Kiran's algorithm to find the value of the last two products."
- 3 minutes: independent work time
- 2 minutes: partner discussion

Activity Synthesis

- Select students to share and explain their calculations of the last two products.
- Consider demonstrating the process of Kiran's strategy to find $3 \times 4{,}132$.
- "Kiran's strategy is called the 'standard algorithm for multiplication.' We'll take a closer look at this algorithm in the next activity."

Activity 2

15 min

Algorithm Comparison

Standards

Addressing **4.NBT.B.5**

The purpose of this activity is for students to compare the standard algorithm for multiplication and an algorithm that uses partial products. The focus of the *Lesson Synthesis* is on the convention used for composing a new unit and how it connects to their work with the standard algorithm for addition.

Access for Students with Disabilities

Engagement: Internalize Self-Regulation. Synthesis: Provide students an opportunity to self-assess and reflect on their own progress. Remind students what kinds of multiplication problems they were working on before this unit, and ask them to notice how the multiplication they are working on now is different. Invite them to list the new strategies they have learned in this section.
Supports accessibility for: Conceptual Processing, Social-Emotional Functioning

Student Task Statement

1. Analyze the two algorithms used to find the value of 4×223.

Launch

- Groups of 2

Activity

- "Work with your partner on the first problem."

Kiran

$$\begin{array}{r} {\scriptstyle 1} \\ 223 \\ \times\quad 4 \\ \hline 892 \end{array}$$

Diego

$$\begin{array}{r} 223 \\ \times\quad 4 \\ \hline 12 \\ 80 \\ +\,800 \\ \hline 892 \end{array}$$

a. How are these algorithms alike? How are they different?

b. Where is the 12 in Kiran's algorithm?

2. a. Try using Kiran's algorithm to find the value of 512×3.

b. Check your work using a different method.

Student Response

1. Sample response:
   a. Alike: They both multiply the same numbers and have the same answer.
   Different: One shows all the place values and the other has a small one on top.
   b. The 12 is decomposed into $10 + 2$. The 2 is in the ones place and the 10 is a 1 added to the tens place.
2. a. 1,536
   b. Students check using an algorithm showing partial products, a diagram, or another method.

- 3–4 minutes: partner work time
- Pause for brief discussion. Invite students to share how the two methods are alike and how they are different.
- "In Kiran's algorithm, why is a 1 written above the 2 in the tens column?" (It represents 1 ten from the number 12. It helps us remember to add it to the tens place because we are doing a lot of calculations in our heads.)
- "Where have we seen this notation before?" (When we add using the standard algorithm, we use this notation to show that we have more than 9 ones, tens or hundreds, and so on, in a given place, we add each group of 10 units to the place value to the left.)
- "Kiran's used the standard algorithm for multiplication. Try using it to find 512×3."
- 3 minutes: independent work time
- 1–2 minutes: partner discussion

Activity Synthesis

- "In the *Lesson Synthesis*, let's compare ways we kept track of the partial products."

Lesson Synthesis

"Today we compared the standard algorithm for multiplication to an algorithm that uses partial-products. Let's see how we'd find 3×512."

Display:

$$\begin{array}{r} 512 \\ \times\quad 3 \\ \hline \end{array}$$

"How would you find the value of the product?" (I know that 500 x 3 is 1500 and 12 x 3 is 36 and 1500 + 36 = 1,536. Three times 2 is 6, so that goes in the ones place. 3 times 1 is 3, so that goes in the tens place. 3 times 5 is 15, so that goes in the thousands and hundreds place.)

Assure students that they are not expected to use a particular method for multi-digit multiplication in IM Grade 4. Explain that they will study this algorithm more in IM Grade 5. Invite them to try to use it to multiply as we continue to work through lessons.

Suggested Centers

- Five in a Row: Multiplication (3–5), Stage 3: Two-Digit Factors (Addressing)
- Compare (1–5), Stage 3: Multiply within 100 (Supporting)

Cool-down

5 min

Choose a Way to Multiply

Standards

Addressing **4.NBT.B.5**

Student Task Statement

Find the value of each product. Explain or show your reasoning.

1. 4×798
2. $8 \times 2{,}864$

Student Response

1. 3,192. Sample responses:
   - $(4 \times 700) + (4 \times 90) + (4 \times 8) = 2{,}800 + 360 + 32 = 3{,}192$
   - I know that 798 is 2 less than 800. So 4 groups of 798 is 4×2 less than 4×800 or 8 less than 3,200, which is 3,192.
2. 22,912. Sample response:
   - $(8 \times 2{,}000) + (8 \times 800) + (8 \times 60) + (8 \times 4) = 16{,}000 + 6{,}400 + 480 + 32 = 22{,}912$

Responding to Student Thinking

Students may decompose the multi-digit factors by place value and correctly find the partial products but make computation errors when adding them.

Next Day Supports
Before the *Warm-up*, invite students to work in small groups to discuss a correct response to this *Cool-down*.